



Dipping Status, Ambulatory Blood Pressure Control, Cardiovascular Disease, and Kidney Disease Progression: A Multicenter Cohort Study of CKD

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Rationale & Objective: Ambulatory blood pressure (BP) monitoring allows concurrent evaluation of BP control and nocturnal BP dipping status, both related to adverse outcomes. However, few studies have assessed the prognostic role of combining information on dipping status and achieved ambulatory BP in patients with chronic kidney disease (CKD).

Study Design: Prospective observational cohort study.

Setting & Participants: 906 patients with hypertension and CKD attending 1 of 3 Italian nephrology clinics.

Exposure: Four groups were defined by simultaneously classifying systolic ambulatory BP levels as being at goal (daytime SBP <135 and nighttime SBP <120 mm Hg) or above goal, and the presence or absence of nocturnal dipping (night-time to daytime SBP ratio of <0.9 versus ≥0.9).

Outcome: The composite of time to initiation of maintenance dialysis or estimated glomerular filtration rate (eGFR) decline ≥50%, and the composite of fatal and nonfatal cardiovascular events.

Analytical Approach: Multivariable Cox proportional hazards models were used to estimate risks of kidney disease progression and cardiovascular disease in the 4 exposure groups where nocturnal dipping with systolic ambulatory BP at goal was the reference group.

Results: The mean patient age was 63.8 years, 61% were male, and 26.4% had diabetes; eGFR was 41.1 ± 20.8 mL/min/1.73 m². The dipping prevalence in each of the 4 groups was as follows: nocturnal dipping with ambulatory BP at goal, 18.6%; no nocturnal dipping with ambulatory BP at goal, 20.5%; nocturnal dipping with ambulatory BP above goal, 11.8%; and no nocturnal dipping with ambulatory BP above goal, 49.1%. Among patients with ambulatory BP above goal, the risk of cardiovascular events was greater in the absence (HR, 2.79 [95% CI, 1.64-4.75]) and presence (HR, 2.05 [95% CI, 1.10-3.84]) of nocturnal dipping. The same held true for risk of kidney disease progression (HRs of 2.40 [95% CI, 1.58-3.65] and 2.11 [95% CI, 1.28-3.48] in the absence and presence of nocturnal dipping, respectively). Patients at the ambulatory BP goal but who did not experience nocturnal dipping had an increased risk of the cardiovascular end point (HR, 2.06 [95% CI, 1.15-3.68]) and the kidney disease progression outcome (HR, 1.82 [95% CI, 1.17-2.82]).

Limitations: Lack of a diverse cohort (all those enrolled were White). Residual uncontrolled confounding.

Conclusions: Systolic ambulatory BP above goal or the absence of nocturnal dipping, regardless of ambulatory BP, is associated with higher risks of cardiovascular disease and kidney disease progression among patients with CKD.

Visual Abstract online

Complete author and article information provided before references.

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Ambulatory blood pressure (BP) monitoring is widely recognized as the gold standard to estimate “true” hypertensive status by identifying BP phenotypes derived from the potential mismatch between ambulatory and

evaluating the prognosis of CKD patients based on ambulatory BP monitoring have focused on the role of nocturnal nondipping status and above-goal ambulatory BP independent of each other even though these 2 conditions are frequently associated. Indeed, current guidelines justify intensifying BP management in patients with above-goal ambulatory BP independent of the patient’s nocturnal dipping status. Conversely, very little is known regarding the management of patients with nondipping status and ambulatory BP at goal. The decision as to whether such patients need to be treated requires as a first step more knowledge of the global risk associated with this phenotype.

Unfortunately, it remains unclear whether the presence of isolated nondipping status (ie, with ambulatory BP at goal) is associated with a different CV risk than that observed in CKD patients with ambulatory BP above goal

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office BP measures. In addition, ambulatory BP is the only tool for assessing abnormal circadian BP patterns such as the absence of physiological BP decline during sleep.¹⁻³

In patients with chronic kidney disease (CKD), ambulatory BP predicts cardiovascular (CV) and kidney disease outcomes better than does office BP.⁴⁻⁹ Similarly, nondipping status is associated with a poor CV and kidney prognosis in CKD patients.⁴⁻⁶ Nonetheless, studies

PLAIN-LANGUAGE SUMMARY

Among patients with chronic kidney disease (CKD), ambulatory blood pressure (BP) monitoring improves the identification of individuals at high risk of clinical disease outcomes. Those with uncontrolled ambulatory BP are known to have a higher risk of developing cardiovascular disease and kidney disease progression, particularly when their ambulatory BP does not decline by at least 10% at night. Whether this is also true for patients with presence of optimal ambulatory BP levels but a BP pattern of no nighttime decline is largely unknown. We measured ambulatory BP in 900 Italian patients with CKD and followed them for several years. We found that, independent of ambulatory BP level, the absence of nighttime reductions in BP was associated with worsening of CKD and more frequent cardiovascular events. The absence of nighttime declines in BP is an independent risk factor for adverse events among patients with CKD. Future studies are needed to examine whether treating the absence of nighttime declines in BP improves clinical outcomes.

and a nondipping pattern. Similarly, the prognostic role of isolated nondipping status on kidney outcome has been investigated only rarely and was limited to CKD patients in Japan.^{10,11} We previously demonstrated that switching administration of 1 antihypertensive drug from morning to evening in CKD patients with nondipping status and 24-hour ambulatory BP at target is effective in restoring physiologic nocturnal BP decline.¹² This finding was not confirmed in Black individuals with hypertensive CKD.¹³ However, no information on outcomes were provided in either study.

Therefore, we designed a prospective observational study to assess the prognostic effect of systolic ambulatory BP goal and dipping status, alone or in combination, on fatal and nonfatal CV events and kidney disease progression in a large cohort of patients with CKD who were not receiving dialysis.

Methods**Study Design and Measurements**

In this multicenter, prospective cohort study, consecutive patients were recruited in 3 nephrology outpatient clinics in 2005-2015. Inclusion criteria were CKD stages 2-5 (not receiving kidney replacement therapy [dialysis or transplant]), hypertension (either office BP $\geq 140/90$ mm Hg without antihypertensive treatment or receiving BP drugs with any BP values), and nephrology care lasting ≥ 6 months with at least 2 visits in the nephrology clinic before enrollment. The exclusion criteria included a recent episode of acute kidney injury (defined as estimated glomerular filtration rate [eGFR] change $>30\%$ in the previous 3 months),

changes in antihypertensive therapy 2 weeks before ambulatory BP monitoring, atrial fibrillation, or inadequate ambulatory BP monitoring (daytime and nighttime systolic/diastolic recordings <20 and <7 , respectively¹⁴). The institutional review boards of the participating centers approved the protocol, and all patients signed informed consent.

Medical history, demographics, laboratory data, and current therapy were collected at baseline. If it was not available from the baseline visit, 24-hour urine collection was requested to measure protein and sodium excretion. Collection was considered inaccurate and was repeated if the value of the measured creatinine excretion rate was outside the 60% to 140% range of the value.⁸ The eGFR was calculated by the 2009 CKD-EPI creatinine equation; because the creatinine was not standardized to isotope-dilution mass spectrometry values, we reduced creatinine levels by 5%.¹⁵

Standardized office BP measurement was obtained during a clinic visit (8:00-11:00 AM), in a sitting position, 3 times at 5-minute intervals by the same physician, who was masked to the results of ambulatory BP recordings to avoid any influence on the office readings. The office BP values were determined as the mean of the 6 values recorded on the 2 consecutive days in which the ambulatory BP monitoring device was installed and removed. Antihypertensive medications were administered from 8:00 AM-10:00 PM to achieve an office BP of $<130/80$ mm Hg (this goal was used during the whole study period).

Ambulatory BP was monitored using a Spacelabs 90207 device with cuff size chosen according to the patient's arm circumference and fixed to the nondominant arm. Three BP readings were taken in the morning (8:00-11:00 AM) concomitant with measurements with an aneroid sphygmomanometer to ensure a difference of less than 5 mm Hg between the 2 sets of values. The monitor recorded BP every 15 minutes between 7:00 AM and 11:00 PM and every 30 minutes between 11:00 PM and 7:00 AM.

Daytime and nighttime periods were derived from the diaries recorded by the patients during the ambulatory BP monitoring, which was always performed on a workday and while the patient was receiving the prescribed antihypertensive therapy. Patients had no access to the ambulatory BP values.

Exposure Factors

To explore the isolated prognostic effect of achieving the target systolic ambulatory BP (daytime systolic BP of <135 mm Hg and nighttime systolic BP of <120 mm Hg) and dipping status (nighttime to daytime systolic BP ratio of <0.9), we classified patients in 4 mutually exclusive categories: nocturnal dipping with ambulatory BP at goal, nocturnal nondipping with ambulatory BP goal, nocturnal dipping with ambulatory BP above goal, and nocturnal nondipping with ambulatory BP above goal.

Outcomes

Co-primary outcomes were the occurrence of fatal and nonfatal CV events and a composite kidney disease

progression outcome. CV events included deaths due to CV causes certified by the National Death Registry and nonfatal events requiring hospitalizations due to myocardial infarction, congestive heart failure, stroke, revascularization procedures, peripheral vascular disease, and nontraumatic amputation. For the CV survival analysis, follow-up was computed from the initial visit to December 2019 or the date of CV event, non-CV death, or first dialysis treatment. The kidney disease progression outcome was a composite of $\geq 50\%$ eGFR decline or initiation of maintenance dialysis. Patients were followed until December 2019 or until the date of the kidney disease progression outcome or death, whichever occurred first.

Statistical Analysis

Continuous variables are expressed as mean and SD or as median and IQR (according to their distribution) and categorical variables were expressed as percentage. Differences between groups were tested by means of 1-way analysis of variance (ANOVA) or Kruskal-Wallis for continuous variables (according to their distribution) and χ^2 test for categorical variables. The median follow-up was estimated by the inverse Kaplan-Meier approach.¹⁶

The incidence rate was determined by dividing the number of events by the total number of person-years accumulated with 95% CI calculated assuming Poisson distribution. Because dialysis and non-CV death before dialysis are competing events for CV outcomes, we calculated the cumulative incidence of fatal and nonfatal CV events by using the competing risk approach described by Fine and Gray.¹⁷ A similar methodology was used for incidence of the kidney disease progression outcome by considering all-cause death as competitive event. Cumulative incidence curves were compared using Gray test.¹⁸

Multivariable Cox proportional hazards models were used to estimate the risk (hazard ratio [HR] and 95% CI) of the CV end point and the kidney disease progression outcome in the 4 groups based on dipping status and ambulatory BP goal (considering nocturnal dipping status and ambulatory BP at goal as the reference group). In the Cox models for the kidney disease progression outcome, death before that outcome was treated as censoring event; in the Cox models for the CV end point, the kidney disease progression outcome and non-CV death were treated as censoring events.

All survival models were stratified by center to take into account potential differences in the baseline risk of the kidney disease progression outcome and the CV end point across the 3 centers. Each model was adjusted for baseline covariates identified a priori: age, sex, body mass index, diabetes, prior CV disease, smoking, left ventricular hypertrophy (LVH) diagnosed by echocardiography, office systolic BP, eGFR, log-transformed proteinuria, total cholesterol, serum phosphate, hemoglobin, use of renin-angiotensin system (RAS) inhibitors, use of statins, and number of antihypertensive drugs.

Two sensitivity analyses were done. First, we included ambulatory BP at goal and dipping status separately or simultaneously in the multivariable Cox models in order to further investigate the isolated effect of ambulatory BP at goal and dipping status on outcomes. Second, we estimated the kidney disease progression risk associated with these 2 variables by using only dialysis initiation as an end point.

Data were analyzed by using STATA 11.2 (StataCorp).

Results

Cohort Characteristics

We included 906 out of 1,116 eligible hypertensive patients. Reasons for exclusion were inadequate ambulatory BP recordings ($n = 65$), recent changes in antihypertensive therapy ($n = 49$), atrial fibrillation ($n = 21$), eGFR change $>30\%$ ($n = 11$), loss to follow-up after baseline visit ($n = 26$), and refusal to participate ($n = 38$).

Overall, nondipping status was detected in 70% of patients, and systolic ambulatory BP was above goal in 61% of the cohort (43% for daytime and 57% for nighttime values). After stratifying by dipping status and ambulatory BP goal, 167 patients (18%) had dipping status with ambulatory BP at goal, 187 (21%) had nondipping status and ambulatory BP at goal, 103 (11%) had dipping status and ambulatory BP above goal, and the remaining 449 (50%) had nondipping status and ambulatory BP above goal.

The baseline demographic and clinical characteristics of these 4 groups are presented in Table 1. All patients were White; those with nocturnal nondipping status with ambulatory BP above goal were older, predominantly male, and had a higher prevalence of diabetes and prior CV disease. They also had more advanced kidney disease as evidenced by lower eGFR and higher proteinuria. Patients with only 1 abnormal parameter (ie, either nondipping status or ambulatory BP above goal) had a risk profile intermediate between the reference group (nocturnal dipping and ambulatory BP at goal) and patients with nondipping status and ambulatory BP above goal. Cause of kidney disease did not differ across the 4 groups (hypertensive nephropathy, 33%; diabetic kidney disease, 22%; glomerulonephritis, 18%; autosomal polycystic kidney disease, 5%; and other/unknown, 22% [$P = 0.2$]).

BP values and treatment in the 4 groups are depicted in Table 2. No difference was found in number of antihypertensive drugs across groups. However, those with nondipping status and ambulatory BP above goal received fewer RAS inhibitors (likely due to a reduced glomerular filtration rate) and more calcium channel blockers, central α -agonists, and adrenergic blockers. Associated with a higher prevalence of prior CV disease and diabetes, prescription of antiplatelet drugs was more frequent in this group while prescription rate of statins was lower. Adherence to low-sodium diet was limited in all groups.

Table 1. Demographic and Clinical Features of Patients Stratified by Achievement of Systolic Ambulatory BP Goal and Dipping Status

	Ambulatory BP at Goal		Ambulatory BP Above Goal		P
	Dippers (n = 167)	Nondippers (n = 187)	Dippers (n = 103)	Nondippers (n = 449)	
Age, y	58.7 ± 15.0	62.2 ± 14.2	65.5 ± 13.1	66.0 ± 13.4	<0.001
Male sex	47.9%	57.8%	68.9%	65.5%	<0.001
Diabetes	19.7%	23.0%	34.0%	28.5%	0.03
History of CV disease	22.8%	25.7%	33.0%	31.6%	0.09
Smoking	23.4%	21.4%	30.1%	26.7%	0.3
LVH	50.9%	59.4%	65.1%	78.4%	<0.001
BMI, kg/m ²	27.6 ± 4.4	28.5 ± 5.4	27.7 ± 4.8	27.7 ± 5.1	0.3
eGFR, mL/min/1.73 m ²	49.9 ± 23.5	43.9 ± 19.2	40.0 ± 22.4	36.7 ± 21.0	<0.001
Cholesterol, mg/dL	195 ± 42	186 ± 40	186 ± 40	193 ± 50	0.1
Calcium, mg/dL	9.5 ± 0.6	9.4 ± 0.6	9.4 ± 0.5	9.4 ± 0.6	0.1
Phosphate, mg/dL	3.9 ± 0.8	3.8 ± 0.8	3.9 ± 0.8	3.9 ± 0.7	0.7
Hemoglobin, g/dL	13.1 ± 1.7	12.9 ± 1.7	12.6 ± 1.8	12.3 ± 1.9	<0.001
Proteinuria, g/d	0.20 [0.08-0.68]	0.25 [0.09-0.83]	0.27 [0.06-1.00]	0.60 [0.11-1.50]	<0.001
24-Hour urinary sodium excretion, mEq/d ^a	146 ± 61	147 ± 55	153 ± 62	155 ± 66	0.4

Continuous data given as mean ± SD or median [IQR]. Ambulatory BP at goal defined as daytime and nighttime SBP of <135 and <120 mm Hg, respectively; dipping status defined as ratio of nighttime to daytime SBP of <0.9. Abbreviations: BMI, body mass index; BP, blood pressure; CV, cardiovascular; eGFR: estimated glomerular filtration rate; LVH, left ventricular hypertrophy; SBP, systolic blood pressure.

^aAvailable in 708 patients.

Table 2. Blood Pressure Characteristics and Therapeutic Features of Patients Stratified by Achievement of Systolic Ambulatory BP Goal and Dipping Status

	Ambulatory BP at Goal		Ambulatory BP Above Goal		P
	Dippers (n = 167)	Nondippers (n = 187)	Dippers (n = 103)	Nondippers (n = 449)	
Office					
Systolic BP, mm Hg	140 ± 19	132 ± 17	153 ± 17	149 ± 19	—
Diastolic BP, mm Hg	82 ± 11	78 ± 10	83 ± 12	83 ± 11	—
24-hour					
Systolic BP, mm Hg	117 ± 8	113 ± 8	140 ± 11	141 ± 14	—
Diastolic BP, mm Hg	71 ± 8	68 ± 8	77 ± 11	78 ± 11	—
Daytime					
Systolic BP, mm Hg	123 ± 8	115 ± 8	147 ± 12	141 ± 15	—
Diastolic BP, mm Hg	75 ± 9	71 ± 8	82 ± 12	79 ± 11	—
Nighttime					
Systolic BP, mm Hg	105 ± 8	110 ± 7	125 ± 11	140 ± 15	—
Diastolic BP, mm Hg	61 ± 7	64 ± 8	67 ± 11	75 ± 11	—
Nighttime to daytime ratio of systolic BP	0.85 ± 0.04	0.96 ± 0.04	0.85 ± 0.04	0.99 ± 0.07	—
BP-lowering drugs					
No. of drugs	2 [1-3]	2 [1-3]	3 [1-3]	3 [1-4]	0.6
ACEI and/or ARB	84.4%	80.8%	76.7%	63.9%	<0.001
Diuretic	35.3%	41.2%	44.7%	36.1%	0.3
Calcium channel blockers	38.9%	41.7%	45.6%	54.6%	0.001
β-Blockers	34.8%	32.1%	37.9%	34.1%	0.8
Central α agonists	6.6%	7.5%	9.7%	17.4%	<0.001
Adrenergic blockers	4.8%	3.7%	1.9%	10.0%	<0.001
Low-sodium diet ^a	26.1%	20.8%	20.7%	22.3%	0.7
Statins	37.8%	38.5%	29.1%	24.1%	<0.001
Epoetin	4.8%	4.8%	10.7%	9.4%	0.1
Antiplatelets ^b	23.6%	26.0%	24.5%	40.8%	<0.001

Continuous data given as mean ± SD or median [IQR]. Abbreviations: ACEI, angiotensin-converting enzyme inhibitors; ARB, angiotensin II receptor blockers; BP, blood pressure.

^aAvailable in 708 patients; low sodium diet defined as urinary sodium excretion ≤100 mEq/d.

^bAntiplatelets include aspirin, clopidogrel, dipyridamole, and ticlopidine.

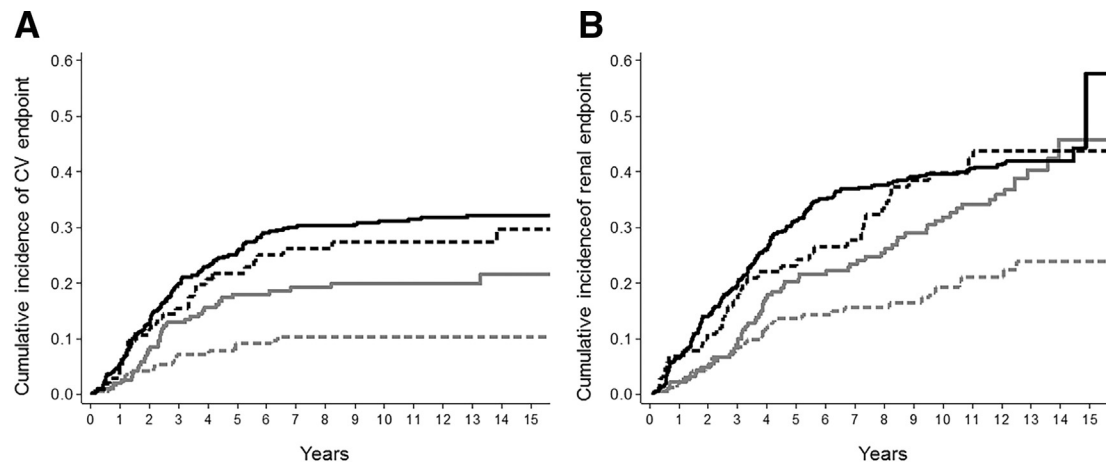


Figure 1. Cumulative incidence of composite CV (A) and kidney disease progression end points (B) by competing risk analysis in chronic kidney disease patients stratified in 4 groups according to dipping status (nighttime to daytime ratio of SBP of <0.9) and ambulatory BP goal (daytime and nighttime SBP <135 and <120 mm Hg, respectively). Patients with nocturnal dipping status with ambulatory BP at goal are depicted by the grey dotted line. Those with nondipping status and ambulatory BP at goal are the grey solid line. Those with dipping status with ambulatory BP above goal are the black dotted line. Those with nondipping status with ambulatory BP above goal are the black solid line. Abbreviation: BP, blood pressure; CV, cardiovascular; SBP, systolic blood pressure.

Survival Analysis: Cardiovascular Risk

During a median follow-up period of 7.8 (IQR, 3.7–12.7) years we recorded 220 CV events (116 nonfatal) corresponding to 3.80 (95% CI, 3.33–4.33) events per 100 patient-years. Competing risk analysis showed progressive greater incidence of the CV end point across the 4 groups, with the highest incidence occurring in patients with nocturnal nondipping status and ambulatory BP above goal (P for trend = 0.001; Fig 1A). After adjustment for potential confounders, the risk for fatal and nonfatal CV events was 2.1-fold higher in patients with nocturnal nondipping status and ambulatory BP at goal, compared with the reference group of dipping status and ambulatory BP at goal (Fig 2). Among patients with ambulatory BP above goal, CV risk was 2.1- and 2.8-fold higher in patients with dipping and nondipping status, respectively. The degree to which risk was greater among patients with ambulatory BP above goal did not differ based on dipping status ($P = 0.2$).

The isolated effect of ambulatory BP at goal and dipping status on the risk of the CV end point persisted when ambulatory BP at goal and dipping status were included separately (HRs of 1.70 [95% CI, 1.20–2.41] and 1.73 [95% CI, 1.23–2.42], respectively); in the model including both variables, ambulatory BP at goal (HR, 1.53 [95% CI, 1.07–2.19]) and dipping status (HR, 1.58 [95% CI, 1.11–2.23]) remained significantly associated with CV outcome.

Survival Analysis: Kidney Disease Progression Outcome

We recorded 315 kidney disease progression events during a median follow-up period of 10.5 (IQR, 4.9–13.4) years, corresponding to a crude event rate of 4.94 (95% CI, 4.42–5.52) per 100 patient-years. Of these patients, 135

experienced an eGFR decline of $\geq 50\%$, and 180 initiated maintenance dialysis. The incidence rate of kidney disease progression events in the 4 groups is depicted in Figure 1B. The highest adjusted risk of this outcome was observed in patients with ambulatory BP above goal, with no difference by nocturnal dipping status ($P = 0.5$). Among patients with ambulatory BP at goal, nondipping status was associated with 82% greater risk of the kidney disease progression outcome (Fig 2).

The isolated effect of ambulatory BP at goal and dipping status on the risk of the kidney disease progression outcome persisted when ambulatory BP at goal and dipping status were included separately (HRs of 1.66 [95% CI, 1.23–2.23] and 1.52 [95% CI, 1.15–2.02], respectively); in the model including both variables, ambulatory BP at goal (HR, 1.52 [95% CI, 1.12–2.07]) and dipping status (HR, 1.39 [95% CI, 1.04–1.86]) remained significantly associated with the kidney disease progression outcome.

In a sensitivity analysis, we defined the kidney disease progression outcome as initiation of maintenance dialysis; this outcome was recorded in 290 patients (180 without eGFR decline of $\geq 50\%$ and 110 after detecting eGFR decline of $\geq 50\%$). The adjusted risk of dialysis initiation was 1.60 (95% CI, 1.01–2.54) in patients with ambulatory BP at goal and nondipping status; among patients with ambulatory BP above goal, the adjusted risk was 1.72 (95% CI, 1.02–2.90) and 2.19 (95% CI, 1.42–3.38) in those with dipping and nondipping status, respectively.

Discussion

In this prospective study performed in over 900 patients with hypertension and CKD, 24-hour BP monitoring

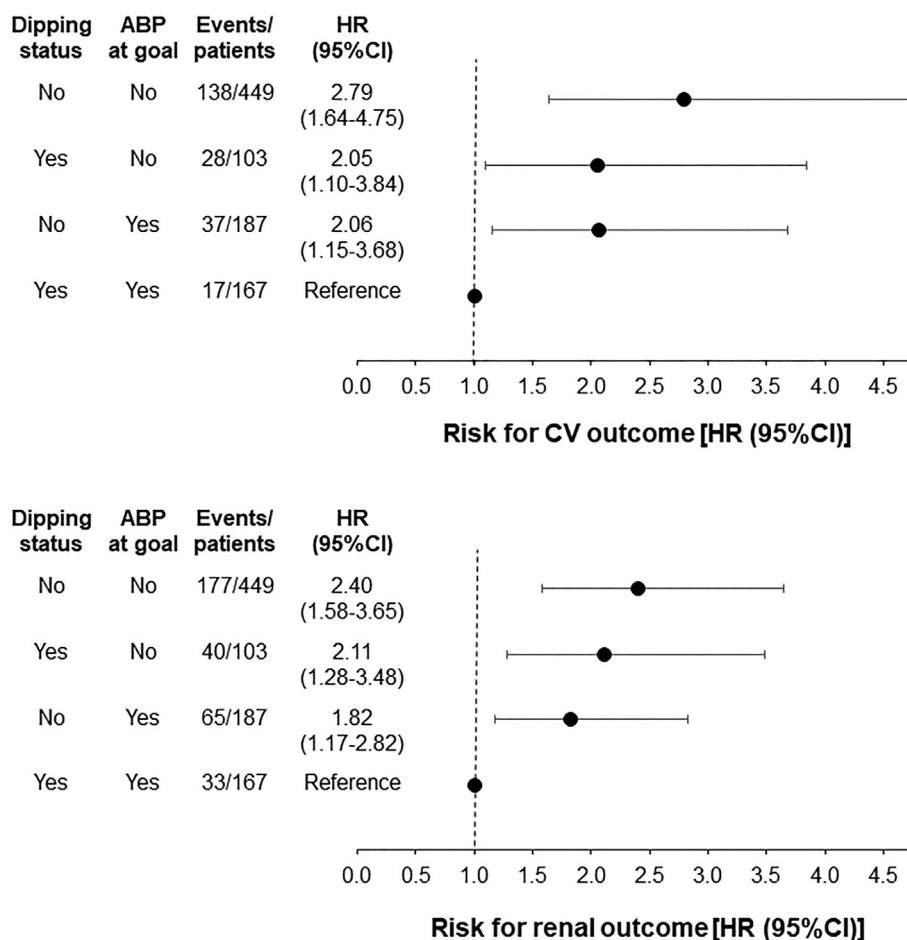


Figure 2. Adjusted HR for composite CV (A) and kidney disease progression end points (B) in chronic kidney disease patients stratified in 4 groups according to dipping status (nighttime to daytime ratio of systolic BP of <0.9) and ABP goal (daytime and nighttime systolic BP <135 and <120 mm Hg, respectively). Abbreviations: ABP, ambulatory blood pressure; BP, blood pressure; CV, cardiovascular; HR, hazard ratio.

demonstrated that the coexistence of an altered circadian BP profile together with systolic ambulatory BP above goal is the most frequent pattern (detected in about half of patients) and is associated with highest risk of CV and kidney disease outcomes. Of major interest, absence of nocturnal dipping constitutes a significant predictor of CV and kidney disease outcomes even when ambulatory BP is at goal.

These findings add novel information on the prognostic relevance of ambulatory BP measurement in the CKD population. Indeed, it is well recognized that ambulatory BP better predicts CV and kidney disease risk in CKD patients as compared with office BP.⁴⁻⁹ Similarly, we know that nondipping is highly prevalent in CKD cohorts and is more frequently observed in older patients as well as in those with more severe kidney disease.¹⁹ This abnormal BP pattern correlates with a greater risk of CV and kidney disease events.⁴⁻⁶ However, the predictive role of ambulatory BP above goal or nondipping status has been always evaluated separately by estimating their effect after adjustment for each other. Because in clinical practice these

conditions frequently coexist, in the current study we combined information on dipping status and ambulatory BP to assess their isolated prognostic effect on CV and kidney disease outcomes.

In our cohort, nondipping status was detected in 70% of patients, while daytime and/or nighttime systolic BP was above goal (135 and 120 mm Hg, respectively) in 61% of the patients. Stratifying patients by these ambulatory BP characteristics is useful to better define prognosis in a CKD population. Indeed, such an approach allows the identification of patients who carry the highest incidence rate of CV and kidney disease events as being those with nondipping status and ambulatory BP above goal (a subgroup constituting half of the cohort). Although the coexistence of ambulatory BP above goal and nondipping status is expected to expose CKD patients to higher risk, it is important to note that each of these alterations per se is associated with an increased CV risk (Fig 2). To our knowledge, this finding has not been previously described in a CKD population. The few studies that have shown that

CV risk is influenced by nocturnal nondipping status in the presence of ambulatory BP at goal were performed in normotensive participants, patients with treated hypertension from the general population,^{20,21} or in a small group of elderly participants with hypertension.²²

The mechanisms underlying the increased CV risk in CKD patients with isolated nondipping status are not readily apparent. Our group previously demonstrated that nighttime but not daytime BP is a stronger independent predictor of concentric and eccentric LVH.²³ In addition, patients with essential hypertension, showing nondipping status and ambulatory BP at goal display higher left ventricular mass index, left atrial volume, and relative wall thickness.^{24,25} Therefore, we can hypothesize that the higher CV risk observed in our CKD patients with nondipping status may be related to cardiac alterations induced by the altered circadian rhythm persisting despite ambulatory BP at goal. Unfortunately, we cannot confirm this hypothesis in our cohort because LVH was collected as a dichotomous variable and no data on either left ventricular mass or cardiac function were available.

Similar to the CV risk findings, having either ambulatory BP above goal or nondipping status exposes patients to a greater risk of faster progression of kidney disease (Fig 2) and dialysis initiation. This finding is consistent with our previous study reporting that the highest risk of death from a kidney disease–related cause is observed in CKD patients with nighttime systolic BP above goal and nondipping status.⁴ In the present study, we extend the prognostic role of an altered circadian rhythm to those patients with ambulatory BP at goal.

Our results are in contrast with 2 recent retrospective analyses of clinical charts of Omihachiman Community Medical Center in Japan, reporting that nondipping status in CKD patients with ambulatory BP at goal was not associated with kidney disease events, defined as eGFR decline over 2 years¹⁰ or composite end point of eGFR decline >40%, kidney failure, or death from kidney disease–related cause.¹¹ Of interest, the Japanese studies had a larger prevalence of nondipping status (83%–86%) than our study and other cohorts of CKD patients,^{6,7,9,26,27} including a prospective cohort study in the same country reporting a prevalence of nondipping of 53%.²⁸ Besides the differences in study design and population, the discrepancy with our results can also be explained based on the use of a different definition of ambulatory BP at goal. The Japanese studies defined ambulatory BP at goal based on 24-hour ambulatory BP values (<130/80 mm Hg) compared with our study in which ambulatory BP was classified “at goal” if both daytime and nighttime systolic BP were on target. Furthermore, Ida et al¹¹ did not report either adjusted risks of kidney disease outcome in the 4 groups or the incidence of each component of the kidney disease outcome, thus making impossible any direct comparison. As regards the potential mechanisms, we can hypothesize that the suboptimal control of BP may

accelerate GFR decline because the transmission of higher pressure load to the glomerular structure may accelerate progression of damage in remnant nephrons of individuals with CKD.²⁹

Isolated nondipping status poses concerns regarding the need to increase antihypertensive treatment in patients with ambulatory BP at goal due to the treatment-related risk of hypoperfusion. However, some therapeutic intervention aimed at restoring dipping status—such as extracellular volume control, inhibitors of RAS, or chronotherapy—should be considered in normotensive patients with nondipping status not only because of their increased cardiorenal risk but also in light of the benefits reported by 4 small trials in 872 CKD patients in terms of CV and renal protection.^{30–33}

It is well recognized that disturbances in the circadian rhythm of natriuresis are associated with nondipping status, particularly in CKD patients among whom salt-sensitive hypertension is more frequent.^{34–37} Consequently, a low-salt diet could be more effective in restoring the circadian rhythm of BP as evidenced by small studies in essential hypertension and CKD.^{37,38} Alternatively, it has been proposed that diuretics could also be effective by restoring the normal pattern of urinary sodium excretion.^{39,40} However, this hypothesis has not been confirmed by a recent randomized trial in patients with advanced CKD in which chlorthalidone in comparison with placebo significantly reduced daytime and nighttime BP but to a similar extent and thus not improving the dipping status.⁴¹ In our study, no difference was found with respect to diuretic use or implementation of low-sodium diet across groups (Table 2).

A further strategy to restore normal dipping status is chronotherapy (ie, switching antihypertensive medication from morning to evening). A meta-analysis of studies evaluating bedtime dosing of antihypertensive drugs has shown that such approach produces a significant reduction in nocturnal BP (−6.3/−3.2 mm Hg) with no significant changes in daytime values (−1.7/−1.1 mm Hg) in comparison with morning administration.⁴² We have previously reported that in CKD patients with nondipping status and ambulatory BP at goal chronotherapy was effective in restoring dipping status in 87.5% of cases and that this strategy was associated with proteinuria reduction.¹² These results were not confirmed in a crossover trial performed in Black CKD patients, possibly due to a more pronounced salt-sensitivity blunting the effect of bedtime dosing.¹³

However, before widely implementing therapeutic strategies targeting nondipping status in the presence of normal ambulatory BP, we should wait for more robust evidence from larger ongoing randomized trials, which include more than 13,500 patients, to definitely prove a reduction of risk of adverse events.^{43–45} From a methodological point of view, our findings could be useful for highlighting that assessment of dipping status should be incorporated into a comprehensive evaluation of BP

control that should not be merely limited to the achievement of BP goal.

The primary limitation of this study is inherent to its observational nature, which forestalls any conclusions regarding cause and effect. Moreover, because all the participants were White, the data may not be generalizable to other racial groups who may differ in terms of prevalence of elevated ambulatory BP and dipping status.⁴⁶ In addition, this analysis is based on only 1 instance of ambulatory BP monitoring, which may lessen the accuracy of risk stratification; however, in CKD patients the diagnosis of abnormal 24-hour BP profile is highly reproducible.⁴⁷

In conclusion, nocturnal nondipping status and ambulatory BP above the goal are distinct predictors of CV and kidney disease outcomes in CKD patients with CKD who are not receiving dialysis. Our results confirm the essential role of ambulatory BP measurement to define accurately the hypertensive burden and circadian BP profile, thus allowing better risk stratification in these high-risk patients.

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Results

Dipping
*Night/day ratio of
systolic BP <0.9*

Composite of eGFR decline >50%
or KRT

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